# Which mode are you in?

Using AI on purpose: what kind of help you ask for, and who checks it.

Mode Fluency · classroom pack · CC BY-SA 4.0

# The calculator

In 1980, the US maths teachers' council recommended calculators "at all grade levels".

The machine took the arithmetic. The curriculum moved up.

NCTM, An Agenda for Action, 1980

# Foundations first, then the tool

Calculators did not hold back maths skills when they were part of teaching and testing (54 studies, 2003).

The one harm signal: sustained use in Grade 4 "appears to hinder" basic skills in average students (79 studies, 1986).

**For which problems, when, and who checks?**

Ellington, JRME, 2003 · Hembree & Dessart, JRME, 1986

# Where the comparison breaks

A calculator is never fluently wrong.

An AI can be: wrong, in the same confident voice it uses when it is right.

# Same AI, different results

| | Practice scores | Exam without AI |
|---|---|---|
| AI that gives answers | about **+48%** | about **17% worse** than no AI |
| AI that gives hints, not answers | about **+127%** | no significant difference from no AI |

Unrestricted answers hurt the exam. A tutor with guardrails largely removed the harm.

About 1,000 high-school maths students, Turkey · Bastani et al., PNAS 122(26), July 2025

# Before you type: what are you expecting?

That expectation is your **stance**.

The same question, "explain photosynthesis", can mean:

- give me the answer to copy
- help me understand it before Friday
- check whether my own explanation holds up

# The Oracle

**You expect:** truth to a hard question.

**How it fails:** confident plausible prose taken as fact.

**One rule:** ask for the uncertainty before the answer. Treat the answer as a claim, not a verdict.

> *"Before you answer, tell me which parts you are least sure about."*

# The Tutor

**You expect:** a path to the answer, not the answer.

**How it fails:** being taught toward a goal you never stated.

**One rule:** state what you want to be able to do afterwards before the first question.

> *"I want to solve these on my own by Friday. Don't tell me the answer yet. What questions should I be asking?"*

# The Mirror

**You expect:** my own thinking, clarified.

**How it fails:** flattery mistaken for reflection.

**One rule:** ask for the structure of your argument back, not for agreement.

> *"List the claims I make, in order, and the evidence for each. Don't tell me if they're good."*

# The Adversary

**You expect:** an attack on my plan, or my draft.

**How it fails:** critique of the wrong thing.

**One rule:** name what you are defending and what would change your mind.

> *"I'm arguing the town should build the bypass, because of air quality. I'd change my mind if the traffic just moved to the villages. Attack that."*

# The Librarian

**You expect:** a synthesis of what is known: what has been said, not what is.

**How it fails:** confident retrieval of what was never said.

**One rule:** check the citation exists and says what is claimed.

> *Five sources on coastal erosion, with author, year and title. Then look each one up.*

# Fluently wrong

One AI-style answer, printed. Your job: find what is wrong, and say how you know.

1. Underline every claim you could check.
2. Check two of them against your textbook or notes.
3. Which stance was the question asked in? What should have been asked first?

# Who checks?

Every piece of work with AI help has a checker, named before you start.

**The checker is never the chat that wrote it.**

- you, against a source, your notes or your own working
- you, by doing one part without help
- your teacher, through the part you do unaided

# The mode log

| Date | What I asked | Which mode | What I checked myself | What I changed |
|---|---|---|---|---|
| 6 Mar | where my steps for Q4 go wrong, no answer | Tutor | worked example, p. 41 | the sign in step 2 |

One line, every time you use AI on an assignment.

Made-up entry, for illustration

# Three assignment modes

**A: No AI.** You do it all yourself.

**B: Tutor only.** Tutor, Mirror, Adversary or Librarian; not the Oracle. It writes nothing you hand in. Keep a mode log.

**C: Full use.** Use AI as the rules allow. Add a mode log, a short report on what the AI did and what you changed, and a part you do without AI.

Every assignment sheet says which mode.

# The one rule to take away

**Name the mode before you type. Then do its one check.**

You may use the calculator. You still sign the exam.